

MODULE-1

BODY DETECTION SYSTEM

CODE USED FOR DEMONSTRATION

```
posedetec.py > ...
1  import cv2
2  import mediapipe as mp
3
4  mp_pose = mp.solutions.pose
5  pose = mp_pose.Pose()
6  mp_draw = mp.solutions.drawing_utils
7  cap = cv2.VideoCapture(0)
8
9  while True:
10     success, frame = cap.read()
11     if not success:
12         break
13     rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
14     results = pose.process(rgb_frame)
15     if results.pose_landmarks:
16         mp_draw.draw_landmarks(
17             frame,
18             results.pose_landmarks,
19             mp_pose.POSE_CONNECTIONS
20         )
21     cv2.imshow("Pose Detection", frame)
22     if cv2.waitKey(1) & 0xFF == ord('q'):
23         break
24 cap.release()
25 cv2.destroyAllWindows()
```

Using OpenCV and MediaPipe, we can access the webcam and use body landmarks to detect the body. The accuracy of the body detection depends on various factors like:

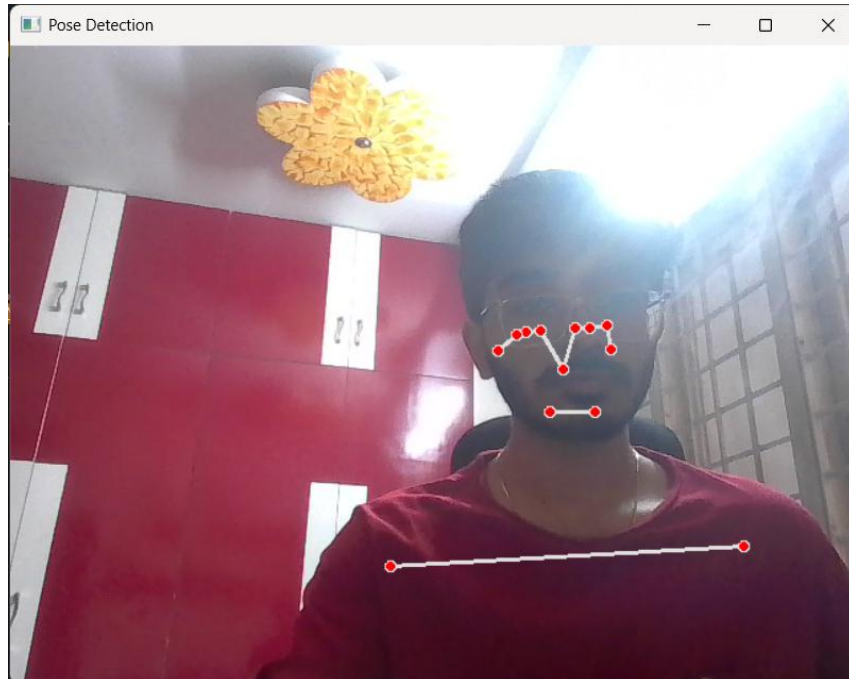
- Lighting
- Body pose
- Proper Camera Distance
- Visibility of the entire body

So, all this factors should be taken into account while trying to get the best results from a body detection system.

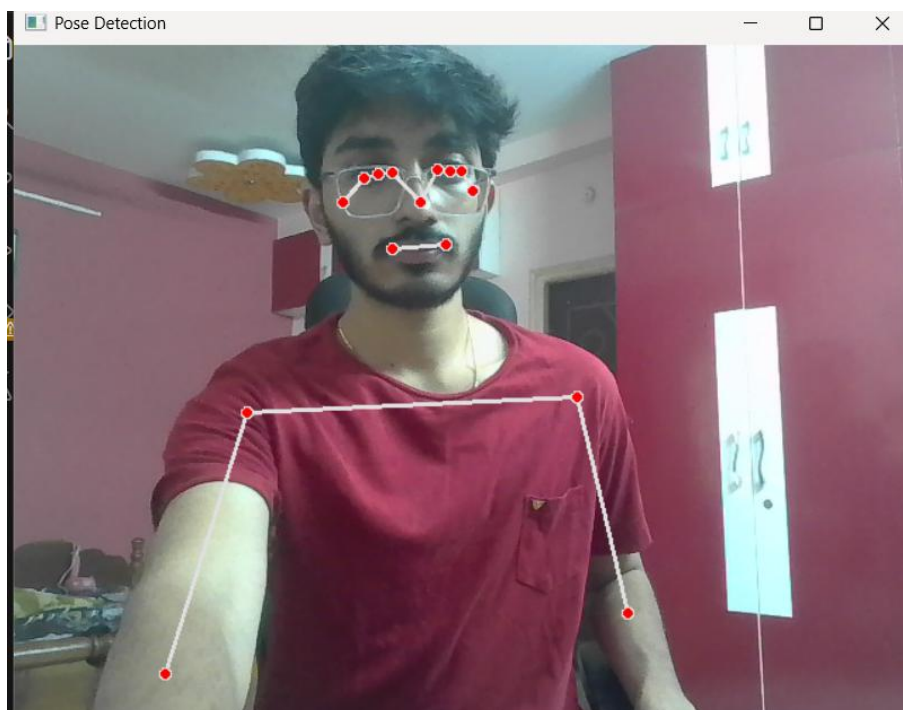
To test and improve its accuracy, it is important to train the model with different poses and lightings as well at different distances from camera to get the best detection.

So, lets check the body detection in different settings:

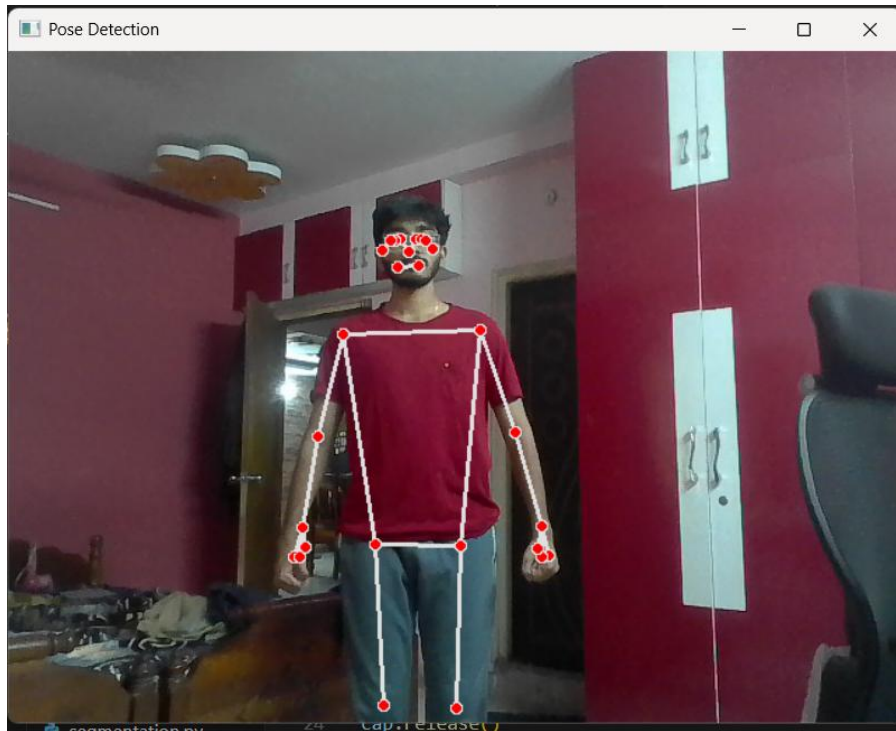
1) **BAD LIGHTING:**



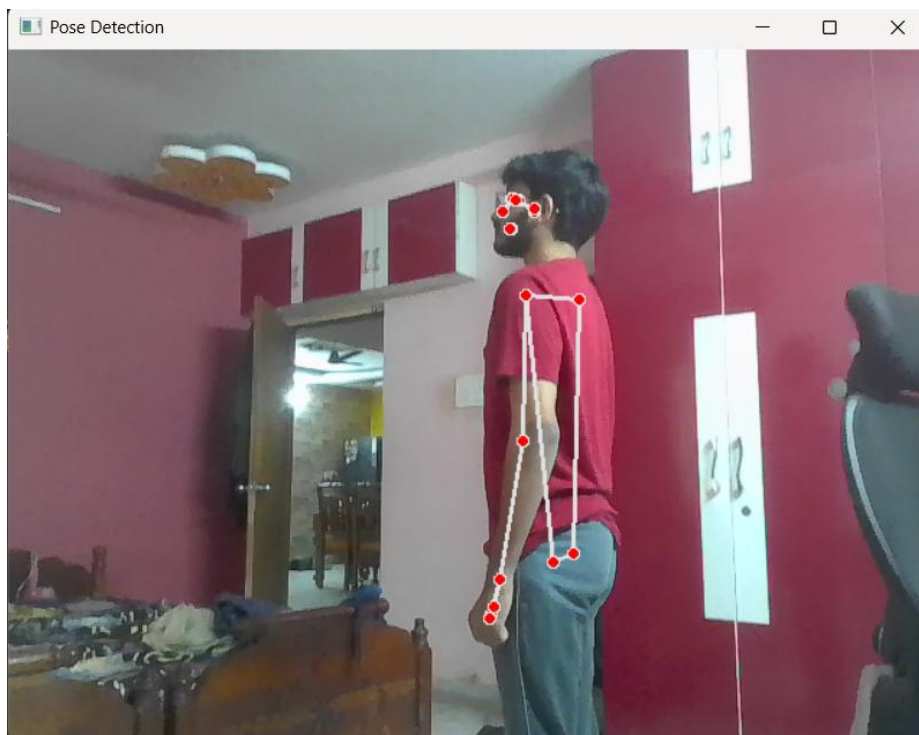
2) **GOOD LIGHTING**



3) STANDING STRAIGHT:



4) STANDING TO YOUR SIDE:



So in conclusion,

As we can see, The Body Detection system is most accurate when the person is standing straight and is in good lighting so this is the basic requirement from the customer to do this for the Raritone app to accurately detect their body entirely for Virtual Try-On and Best Size recommendation.

For the almost perfect body detection, you need other physical hardware like multiple cameras and deep camera that cannot be achieved by an mobile application that is helping with Virtual Try-On through your web or PC camera.

We have to understand that the application we are building won't provide the measurements right to its millimeters but it will be accurate enough to use for Virtual Try-On so the customer can experience how a clothing looks on them and which size is perfect for their body size.